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(54) **CHARGING SYSTEM WITH SERVICE VEHICLE**

(71) Applicant: **Oshkosh Corporation**, Oshkosh, WI (US)
(72) Inventors: **Robert Messina**, Oshkosh, WI (US); **Shashank Bhatia**, Oshkosh, WI (US); **Ignacy Puzskiewicz**, Hagerstown, MD (US); **Mike Buhidar**, Oshkosh, WI (US); **David Lombardo**, Oshkosh, WI (US); **Peter Gilbert**, Oshkosh, WI (US)

(73) Assignee: **OSHKOSH CORPORATION**, Oshkosh, WI (US)

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CPC **B60L 53/36** (2019.02); **B60L 53/12** (2019.02); **B60L 53/38** (2019.02)

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None

See application file for complete search history.

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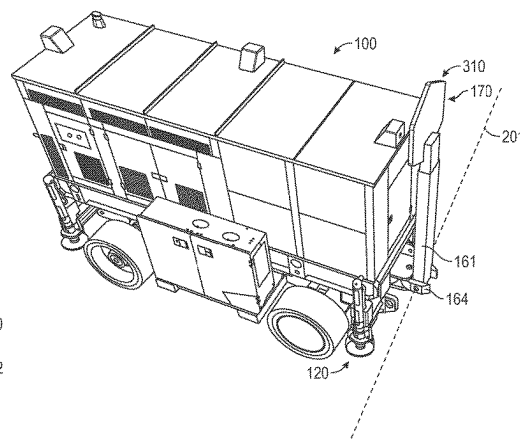
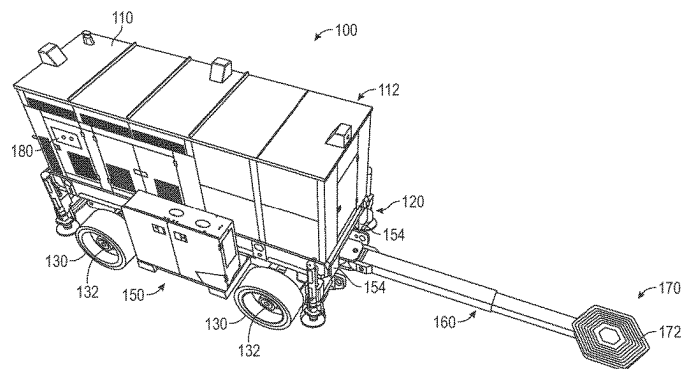
(74) *Attorney, Agent, or Firm* — FOLEY & LARDNER LLP

(57)

ABSTRACT

A charging system includes a service vehicle and a controller. The service vehicle includes a chassis, a series of tractive elements coupled to the chassis, an energy storage device, and a charging interface operatively coupled to the energy storage device. The controller is configured to identify a location of a vehicle, control the service vehicle to navigate to the location of the vehicle, and control the service vehicle to transfer energy from the energy storage device to the vehicle through the charging interface.

7 Claims, 8 Drawing Sheets



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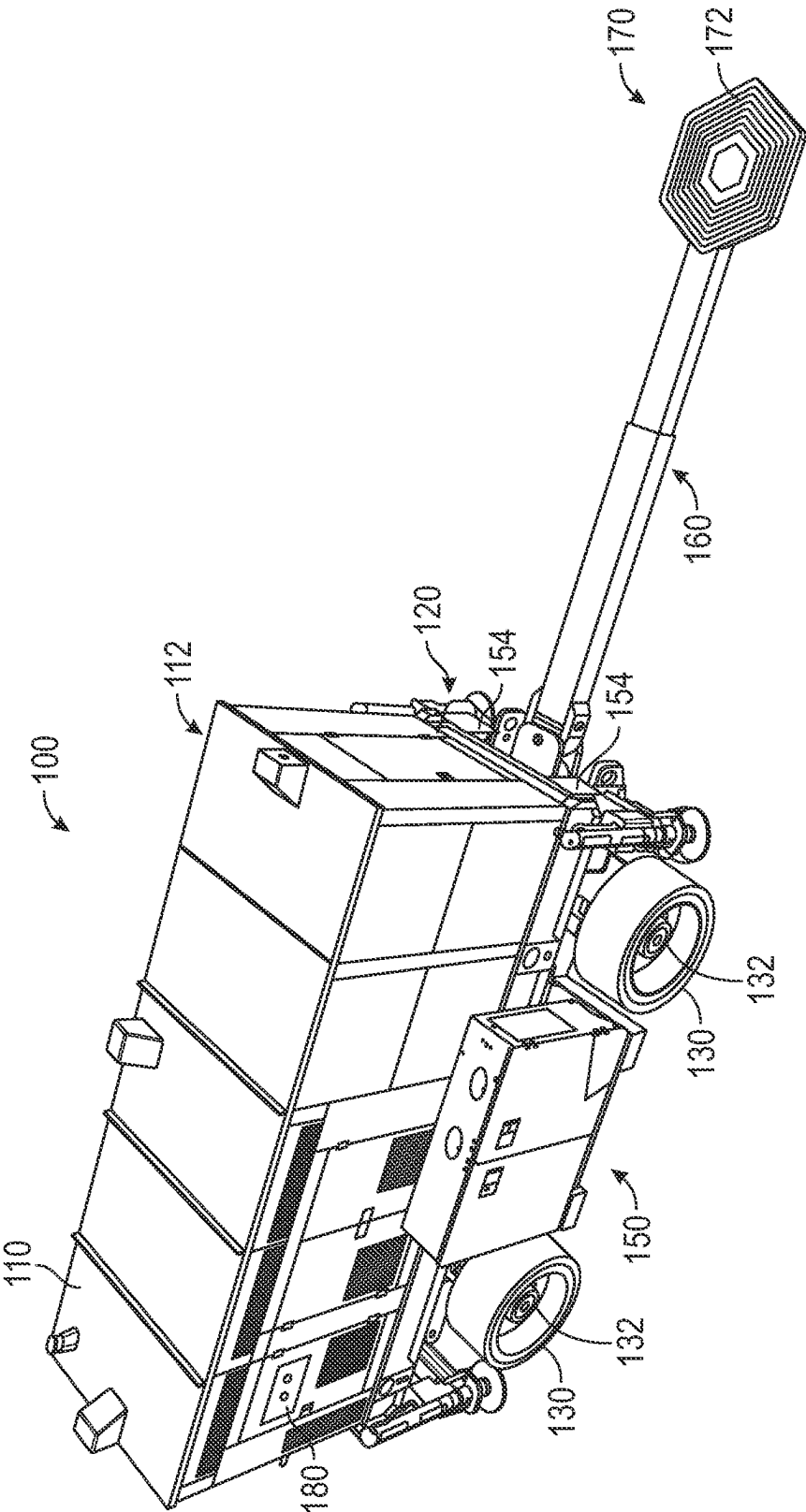


FIG. 1

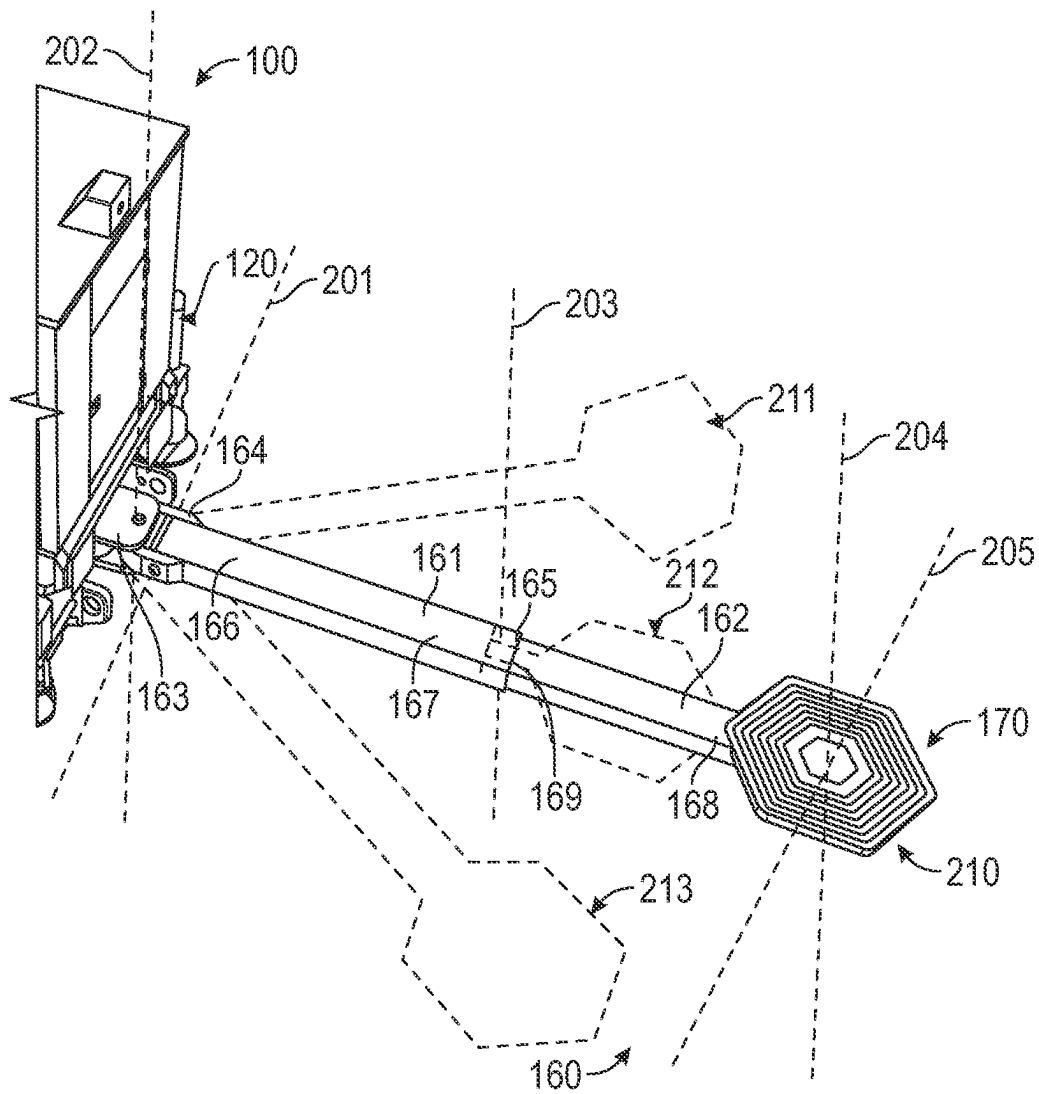


FIG. 2

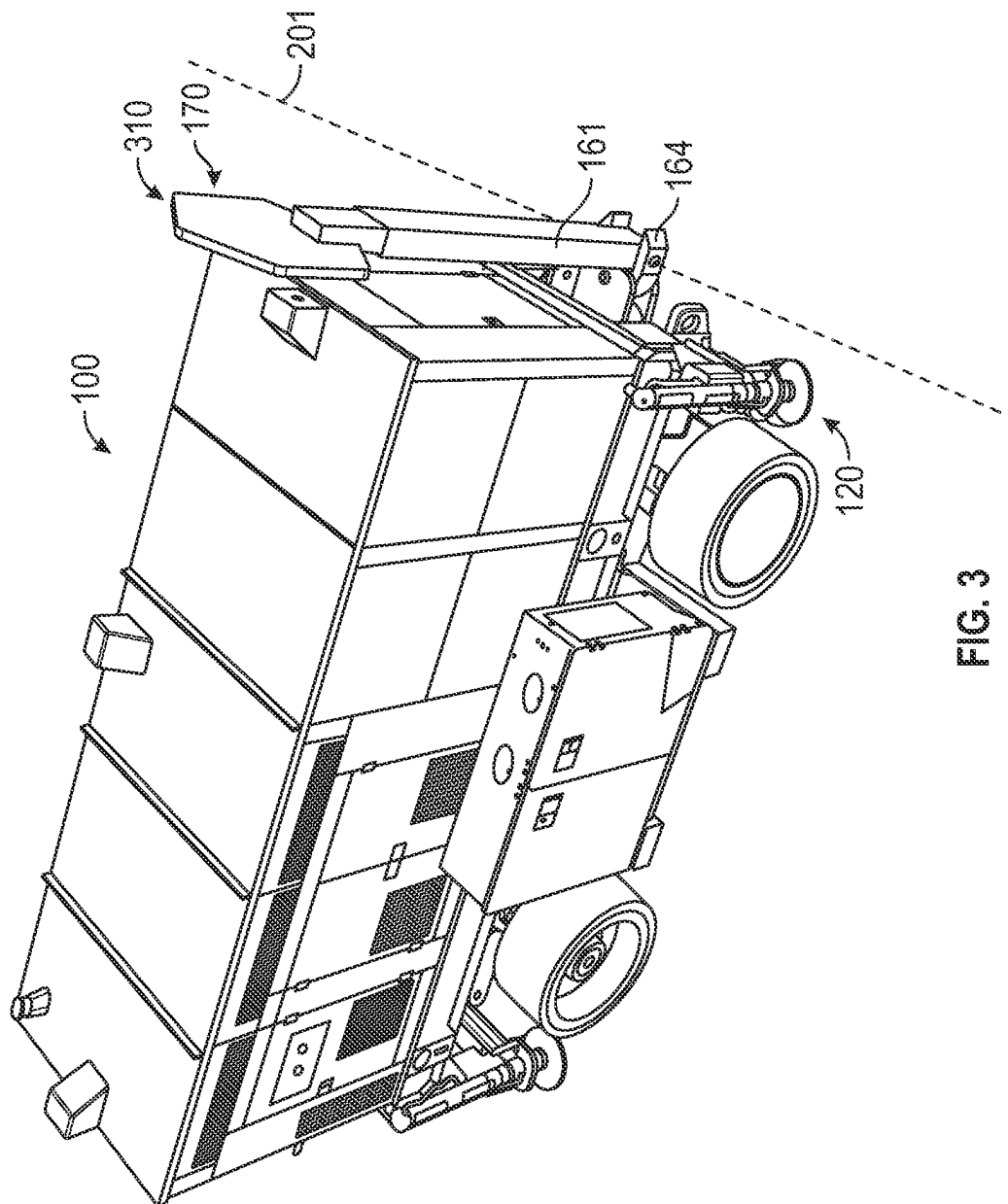


FIG. 3

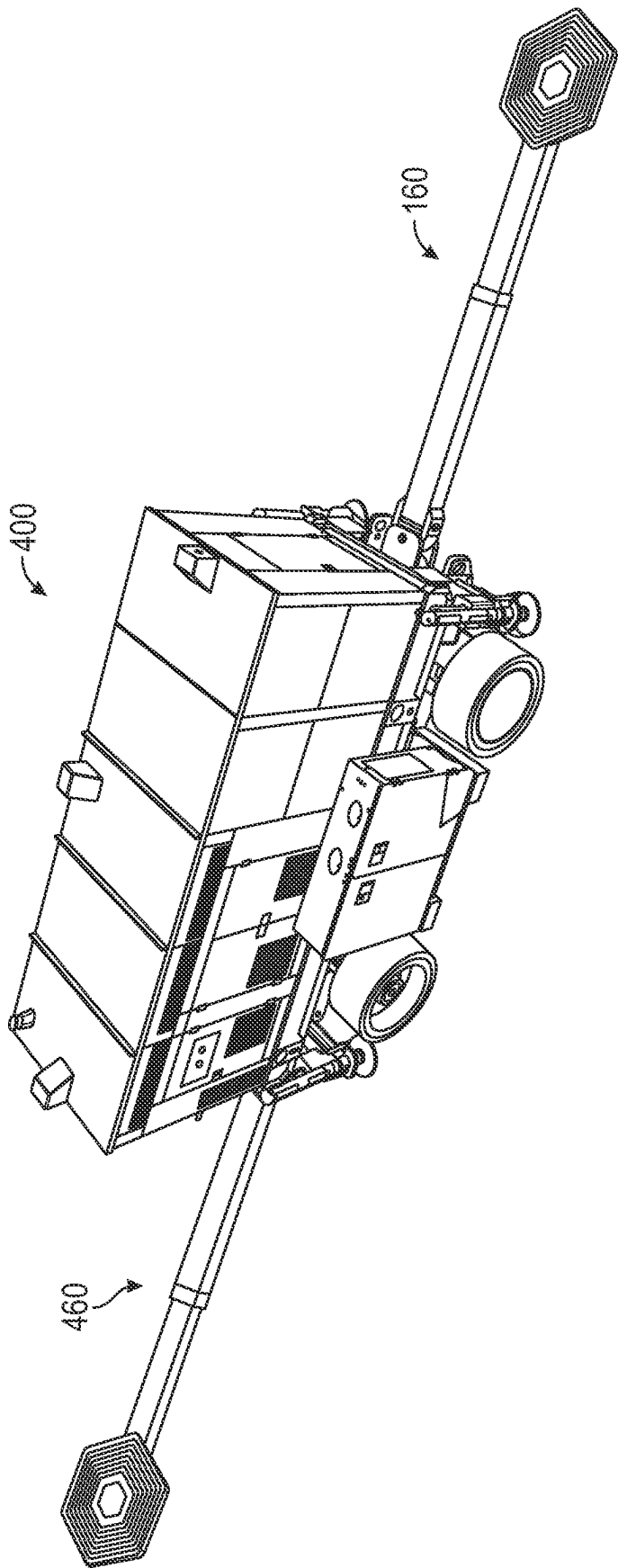


FIG. 4

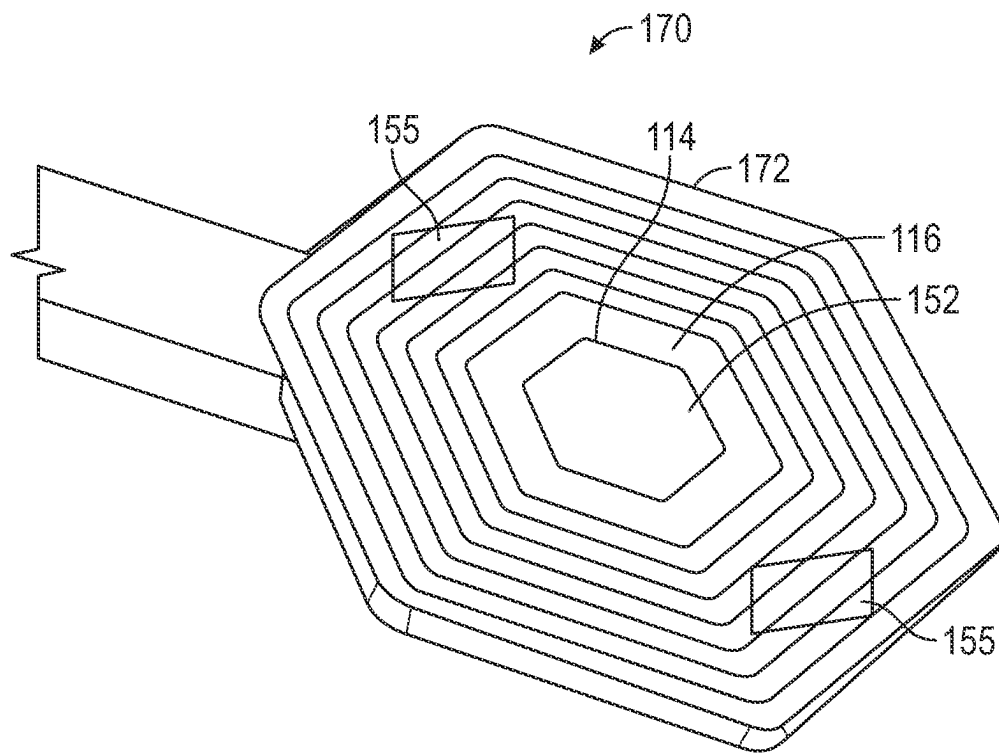


FIG. 5

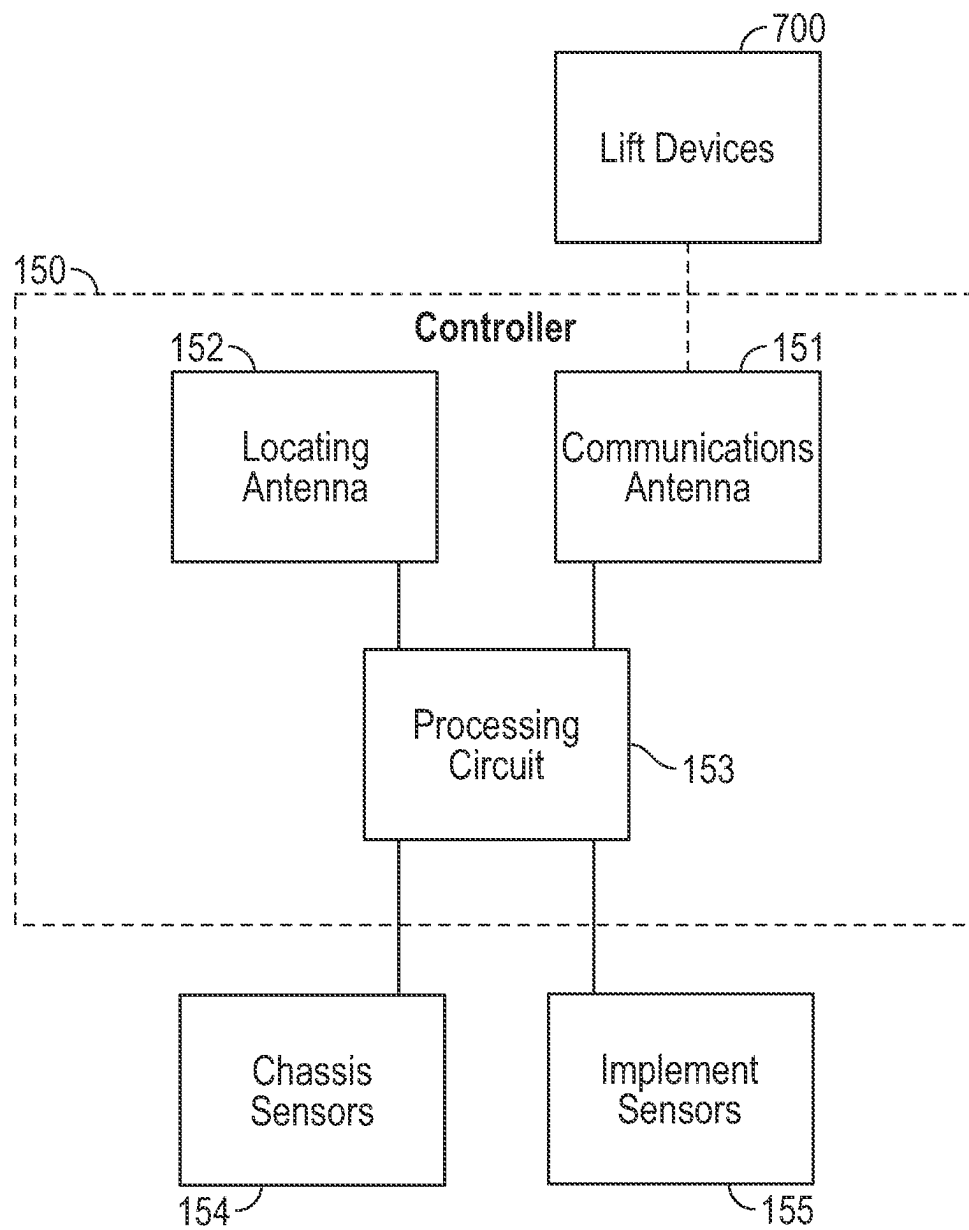


FIG. 6

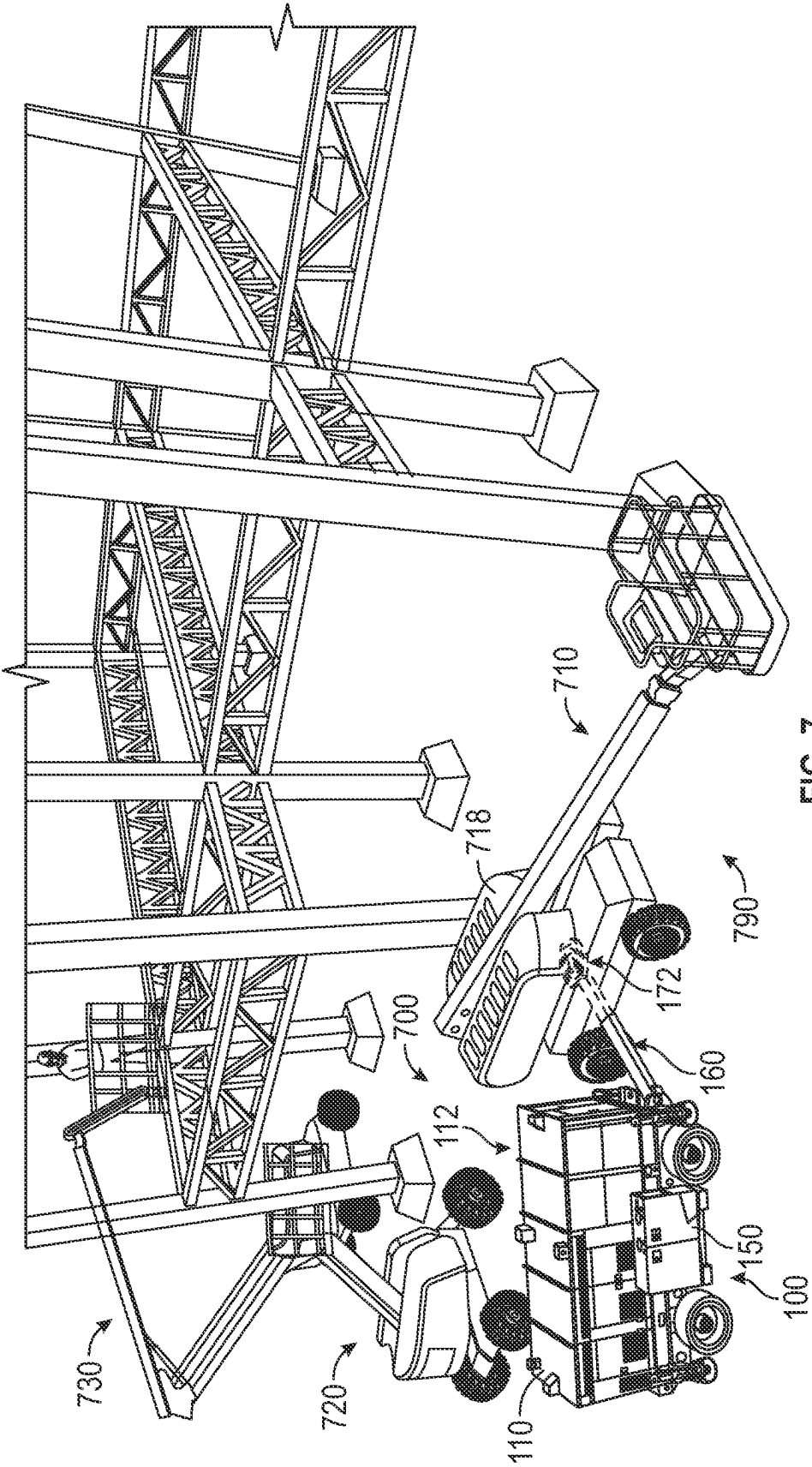


FIG. 7

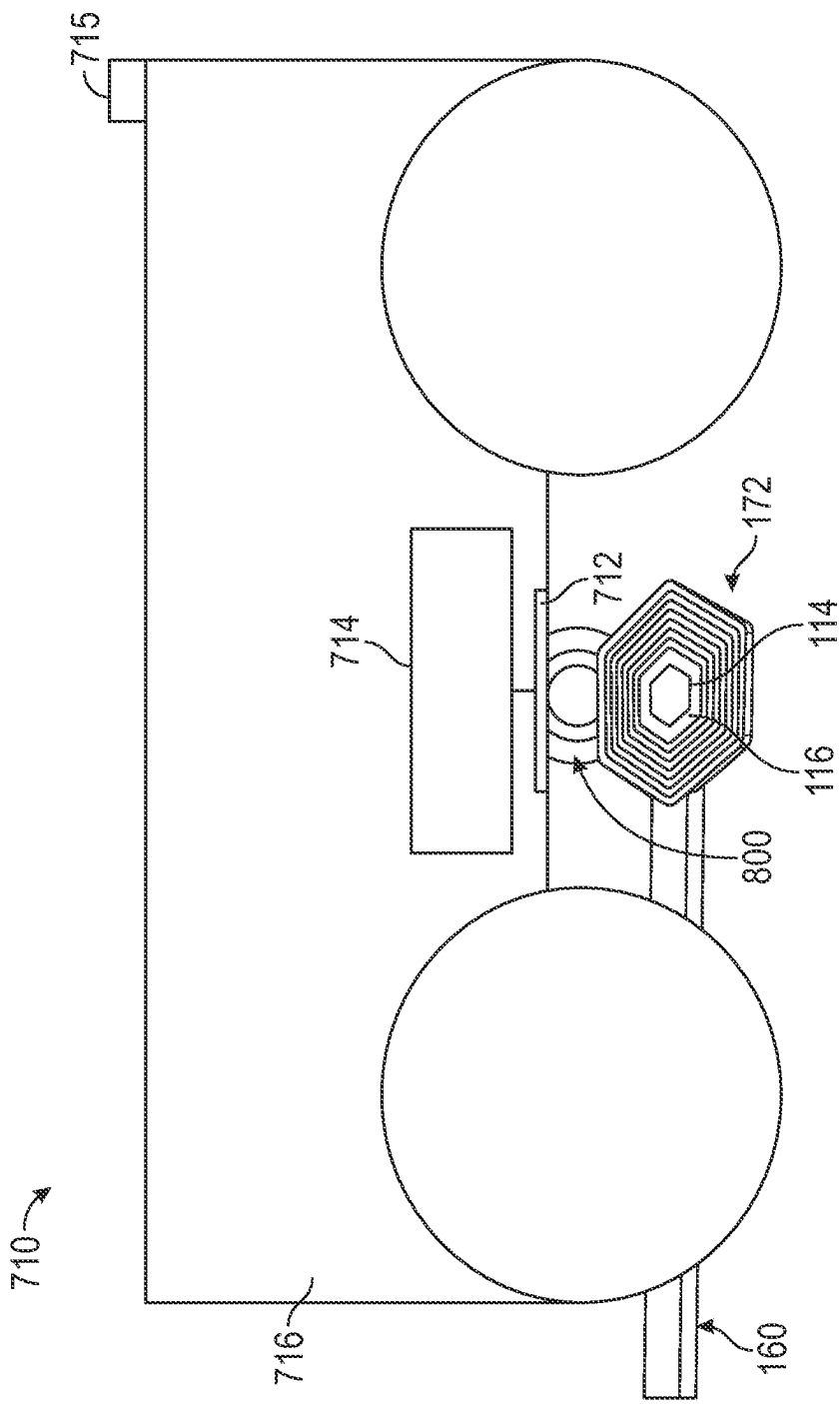


FIG. 8

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CHARGING SYSTEM WITH SERVICE VEHICLE

CROSS-REFERENCE TO RELATED PATENT APPLICATION

This application claims the benefit of and priority to U.S. Provisional Application No. 63/302,184, filed on Jan. 24, 2022, the entire disclosure of which is hereby incorporated by reference herein.

BACKGROUND

Aerial work platforms (AWPs) and mobile elevating work platforms (MEWPs) are increasingly transitioning to semi-electric or all electric configurations. To support the increasing electrification of these AWPs and MEWPs, the vehicles are equipped with one or more charge storing devices, such as batteries. Because the capacity of charge storing devices is limited, recharging is frequently needed.

SUMMARY

At least one embodiment relates to a charging system including a service vehicle and a controller. The service vehicle includes a chassis, a series of tractive elements coupled to the chassis, an energy storage device, and a charging interface operatively coupled to the energy storage device. The controller is configured to identify a location of a vehicle, control the service vehicle to navigate to the location of the vehicle, and control the service vehicle to transfer energy from the energy storage device to the vehicle through the charging interface.

Another embodiment relates to a service vehicle including a chassis, a series of tractive elements coupled to the chassis, a battery pack, a wireless charging interface configured to wirelessly transfer energy from the battery pack to a recipient vehicle, and a boom assembly coupling the wireless charging interface to the chassis and configured to move the wireless charging interface relative to the chassis.

Another embodiment relates to a non-transitory computer-readable medium having instructions stored thereon that, when executed by one or more processors, cause the one or more processors to implement operations. The operations include receiving location data indicating a location of a vehicle to be charged, controlling a service vehicle to autonomously navigate to the location, controlling an actuator assembly of the service vehicle to adjust a position of a wireless charging interface of the service vehicle relative to a chassis of the service vehicle, and controlling the wireless charging interface to transfer energy from a battery pack of the service vehicle to the vehicle to be charged.

This summary is illustrative only and is not intended to be in any way limiting. Other aspects, inventive features, and advantages of the devices or processes described herein will become apparent in the detailed description set forth herein, taken in conjunction with the accompanying figures, wherein like reference numerals refer to like elements.

BRIEF DESCRIPTION OF THE FIGURES

The disclosure will become more fully understood from the following detailed description, taken in conjunction with the accompanying figures, wherein like reference numerals refer to like elements, in which:

FIG. 1 is a perspective view of a service vehicle, according to one embodiment;

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FIG. 2 is a perspective view of the boom assembly of FIG. 1, according to one embodiment;

FIG. 3 is a perspective view of the service vehicle of FIG. 1, showing the boom assembly in a vertical position, according to one embodiment;

FIG. 4 is a perspective view of a service vehicle with a secondary boom assembly, according to one embodiment;

FIG. 5 is a perspective view of the charging pad of the service vehicle depicted in FIG. 1, according to one embodiment;

FIG. 6 is a block diagram showing the controller of the service vehicle depicted in FIG. 1, according to one embodiment;

FIG. 7 is a perspective view of the service vehicle of FIG. 1, shown interacting with a lift on a job site, according to one embodiment;

FIG. 8 is a schematic view of the lift device of FIG. 7 positioned in proximity with the charging pad of FIG. 1, according to one embodiment.

DETAILED DESCRIPTION

Before turning to the figures, which illustrate the exemplary embodiments in detail, it should be understood that the present application is not limited to the details or methodology set forth in the description or illustrated in the figures. It should also be understood that the terminology is for the purpose of description only and should not be regarded as limiting.

Referring to the figures generally, the various exemplary embodiments disclosed herein relate to systems, apparatuses, and methods for charging a lift device, such as an AWP or MEWP. The system for charging a lift generally includes an autonomous service vehicle. The service vehicle includes a chassis, a series of tractive elements coupled to the chassis, an electrical cabinet coupled to the chassis, a battery assembly coupled to the chassis, a boom assembly coupling a charging implement to the chassis, and a controller. The boom assembly includes one or more telescoping sections and one or more actuators configured to move each individual section relative to one another, providing control over the extension of the boom assembly.

The charging implement includes an induction coil (e.g., a copper coil, etc.) that is configured to receive current from an electrical power source, such as a utility source (e.g., from a wall socket, etc.), generator, or battery assembly. When the induction coil is powered, current is supplied from the electrical power source to the induction coil, which creates a magnetic field. The magnetic field extends upwardly and outwardly from the charging implement, such that a lift or other equipment positioned proximate to the charging implement can interact with the generated magnetic field. If the lift or other equipment includes an antenna loop (e.g., a copper coil) in communication with its battery or battery assembly, a current will be generated within the antenna loop when the antenna loop is positioned within the magnetic field generated by the induction coil. The current within the antenna loop can then be supplied to the battery or battery assembly within the lift or other equipment to charge the battery or battery assembly without the need for a wired connection. After a sufficient charge level is achieved, the lift or other equipment can drive away from or otherwise be removed from the charging implement.

Referring now to FIG. 1, a service vehicle **100** is depicted. The service vehicle **100** generally includes a frame, shown as chassis **120**, a series of tractive elements **130** (e.g., wheel and tire assemblies, crawler tracks, etc.) coupled to the

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chassis **120**, an electrical cabinet **110** coupled to the chassis **120**, a battery assembly **112** (e.g., a battery pack, a battery module, etc.) coupled to the chassis **120**, a boom assembly **160** coupling a charging implement **170** to the chassis **120**, and a controller **150** coupled to the chassis **120**. The charging implement **170** includes a charging pad **172**.

The tractive elements **130** engage a support surface to support the service vehicle **100**. In some embodiments, one or more of the tractive elements **130** are driven to propel the service vehicle **100**. As shown, the service vehicle **100** includes a series of actuators or drivers, shown as electric motors **132**, that drive the tractive elements **130**. The electric motors **132** may be powered using electrical energy from the battery assembly **112**. The electric motors **132** may be coupled to a controller, such as the controller **150** depicted in FIG. 1, providing the controller **150** with control over navigation (e.g., steering and propulsion) of the service vehicle **100**.

In some embodiments, the electrical cabinet **110** can support a variety of different electrical components, including transformers that are configured to step down and/or step up voltage received from a secondary source. In some examples, the electrical cabinet **110** also receives one or more inverters. The inverters are configured to transition direct current electricity stored within one or more batteries included in the battery assembly **112** into alternating current electricity for use by the charging pad **172**, as discussed below. The electrical cabinet **110** may be coupled to the controller **150** to communicate data regarding the battery life of the battery assembly **112**.

In some embodiments, the electrical cabinet **110** stores or is coupled to an electrical power source. For example, in some embodiments, the electrical cabinet is placed in communication with a utility source. The utility source can supply standard utility alternating current electrical power at 120 V and 60 Hz, for example. In other embodiments, the electrical cabinet **110** is placed in communication with a 240 V or 480 V power source instead. Additionally or alternatively, the electrical cabinet **110** can support one or more batteries included in the battery assembly **112**. In some examples, a plurality of rechargeable batteries are included in the battery assembly **112** (e.g., lithium-ion, nickel-cadmium, lead-acid, etc.) and are received within the electrical cabinet **110**. The battery assembly **112** is configured to receive electricity from the utility source through the inverter, which converts the AC utility source power into DC power which can be stored within the battery. In some examples, the battery assembly **112** can also be charged from other equipment engaged with the charging pad **172**. In some embodiments, the electrical cabinet **110** houses an internal combustion engine and a generator that are configured to produce and supply power as a secondary power source. In some embodiments, the electrical cabinet **110** is in communication with one or more solar panels that supply electrical energy in response to exposure to light.

In some embodiments, the electrical cabinet **110** is electrically coupled to the charging pad **172** and is configured to supply electrical current to the charging pad **172**. Electrical current is provided from the electrical cabinet **110** to the charging pad **172** by one or more of the power sources in communication with the electrical cabinet **110**. In some examples, a wired connection of one or more cables and/or a plug is formed between the electrical cabinet **110** and the charging pad **172** so that electrical current can be efficiently transmitted between the electrical cabinet **110** and the charging pad **172**. The electrical power transmitted from the electrical cabinet **110** can be preconditioned depending on

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the electrical supply source. For example, electrical power supplied to the charging pad **172** by the utility source can be passed through a transformer before being supplied to the charging pad **172**. Alternatively, electrical power provided from the battery assembly **112** can be passed to an inverter before being supplied to the charging pad **172**, such that alternating current is always provided to the charging pad **172**.

Referring now to FIG. 2, the boom assembly **160** (e.g., an actuator assembly) of the service vehicle **100** depicted in FIG. 1 is shown, according to some embodiments. The boom assembly **160** includes a boom base **161** (e.g., a base boom section), a telescope section **162** (e.g., a fly boom section), a first vertical joint **163**, a second vertical joint **165**, and a horizontal joint **164**. A proximal end **166** of the boom base **161** is pivotably coupled to the chassis **120** by the first vertical joint **163** and the horizontal joint **164**, facilitating rotation of the boom base **161** relative to the chassis **120** about a first horizontal axis **201** and a first vertical axis **202**, respectively. The telescope section **162** is received within the boom base **161** (e.g., the boom base **161** is a hollow, tubular member) and is displaceable (e.g., selectively repositionable) between a retracted position and an extended position. The telescope section **162** is received within a distal end **167** of boom base **161** and is configured to extend and retract, moving the charging implement toward or away from the chassis **120**. The rotation of the boom base **161** about the first vertical axis **202** and the displacement of the telescope section **162** between extended and retracted positions permits lateral and longitudinal movement of the charging implement **170** to reach various positions, including but not limited to positions **210**, **211**, **212**, and **213** of FIG. 2. The distal end **167** of the boom base **161** includes a second vertical joint **165**. When the telescope section **162** is in a fully extended position (e.g., position **210**), a proximal end **169** of the telescope section **162** engages the second vertical joint **165** and may rotate about a second vertical axis **203** relative to the boom base **161**. The articulation of the boom assembly **160** about the second vertical axis **203** may facilitate navigating the charging implement **170** around obstacles (e.g., around the wheels of another vehicle to be charged).

In some embodiments, the charging implement **170** is coupled to the service vehicle **100** by an articulating boom lift (e.g., an articulating boom lift of the lift device **730** shown in FIG. 7), instead of the boom assembly **160** as described above, and the charging implement **170** is coupled to an implement on the end of the articulating boom lift. The articulating boom lift may control the position of the charging implement **170** by manipulating one or more telescopic sections, the one or more telescopic sections coupled to one another by a series of joints, and controlling the positions of the one or more telescopic sections relative to one another with a series of hydraulic assemblies. In this way, the service vehicle **100** may raise and lower the charging implement **170** differently than, or in addition to, as described above in relation to the boom assembly **160**.

In some embodiments, the charging implement **170** is fixedly coupled to a distal end **168** of the telescope section **162**. In other embodiments, the charging implement **170** is rotatably coupled to the distal end **168** of the telescope section **162** by a turntable, allowing the charging implement **170** to rotate relative to the telescope section **162** about a third vertical axis **204**. In other embodiments still, the charging implement **170** may be coupled to the distal end **168** of the telescope section **162** by a second vertical joint,

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facilitating rotation of the charging implement **170** relative to the telescope section **162** about a second horizontal axis **205**.

Movement of the boom assembly **160** may be controlled by one or more actuators. In some embodiments, the actuators are electric actuators (e.g., electric linear actuators) powered by electrical energy from the battery assembly **112**. The boom assembly **160** may include one or more first actuators configured to translationally move the telescope section **162** relative to the boom base **161**. The boom assembly may include one or more second actuators coupled to the first vertical joint **163** and one or more third actuators coupled to the horizontal joint **164**. The second actuators and the third actuators may be configured to control rotation of the boom base **161** relative to the chassis **120** about the first horizontal axis **201** and the first vertical axis **202**, respectively. The boom assembly **160** may further include one or more fourth actuators coupled to the turntable that couples the charging implement **170** to the distal end **168** of the telescope section **162**. The fourth actuators may be configured to control rotation of the charging implement **170** relative to the telescope section **162** about the second vertical axis **204**. The boom assembly **160** may further include one or more fifth actuators coupled to the vertical joint that couples the charging implement **170** to the telescope section **162**. The fifth actuators may be configured to control rotation of the charging implement **170** relative to the telescope section **162** about the second horizontal axis **205**. The first, second, third, fourth, and/or fifth actuators of the boom assembly **160** may be coupled to a controller, such as the controller **150** depicted in FIG. 1, providing the controller **150** with control over the position of the charging implement **170**.

Referring now to FIG. 3, a perspective view of the service vehicle **100** depicted in FIG. 1 is shown, according to one embodiment. As described in relation to FIG. 2, the boom base **161** may rotate relative to the chassis **120** about the first horizontal axis **201**, moving the charging implement **170** to reach various positions including, but not limited to, position **310**. In the position **310**, the boom base **161** is oriented substantially vertically, adjacent the electrical cabinet **110**. This reduces the footprint of the service vehicle **100** relative to an active position (e.g., the positions **210**, **211**, **212**, and **213** of FIG. 2) that places the charging implement **170** farther from the electrical cabinet **110** and the chassis **120**. The position **310** may act as a storage position or transport position, facilitating unobstructed movement of the service vehicle **100**. In the active position, the charging implement **170** may face upward (e.g., as shown in FIG. 2) to interact with an antenna coil **712** along a bottom surface of a lift device **700**. Alternatively, the charging implement **170** may rotate about the horizontal axis **205** to face longitudinally outward, away from the chassis **120**, to interact with an antenna coil **712** along a front, rear, left, or right side of a lift device **700**. Alternatively, the charging implement **170** may rotate about the horizontal axis **205** to face downward to interact with an antenna coil **712** along a top side of a lift device **700**.

Referring now to FIG. 4, a perspective view of a service vehicle **400** is shown, according to one embodiment. The service vehicle **400** may be substantially similar to the service vehicle **100**, except the service vehicle **100** further includes a secondary boom assembly **460**. Accordingly, any description with respect to the service vehicle **100** may also apply to the service vehicle **400**. The secondary boom assembly **460** may be substantially similar to the boom assembly **160**. Accordingly, the secondary boom assembly

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460 may operate similarly to the boom assembly **160**. In other embodiments, the secondary boom assembly **460** may be configured to operate differently than the boom assembly **160**. By including two boom assemblies, the service vehicle **400** may charge two or more external vehicles simultaneously.

Referring now to FIG. 5, a perspective view of the charging implement **170** of the service vehicle **100** depicted in FIG. 1 is shown, according to one embodiment. Electrical current received by the charging pad **172** is routed to an induction coil **114** positioned within the charging pad **172**. The induction coil **114** includes a wire structure formed of a conductive material (e.g., copper) having one or more turns or coils. When an electrical current is provided to the induction coil **114**, the current travels through the wire structure in a circular manner. Movement of the current through the induction coil **114** creates a magnetic field that extends vertically upward, through the upper surface **152** of the charging pad **172** and above the charging pad **172**, generally. The magnetic field generated by the induction coil **114** can then be used to generate and wirelessly charge lifts and other equipment positioned within the magnetic field, as explained in additional detail below.

In some embodiments, the induction coil **114** is positioned within a charging area **116** formed within the charging pad **172**. The charging area **116** may represent a range of locations within which the magnetic field is strongest, and thus charging is most effective. As depicted in FIG. 1, the charging area **116** can be visually marked on the charging pad **172** (e.g., with different coloration, with a series of concentric rings, etc.) so that an operator of a lift device or other vehicle can easily identify the location in which the induction coil **114** is positioned. In some examples, the charging area **116** is centrally located within the charging pad **172**. In other examples, the charging area **116** is offset to one side of the charging pad **172**.

Referring now to FIG. 6, a block diagram of the controller **150** of the service vehicle depicted in FIG. 1 is shown, according to one embodiment. The controller **150** includes a processing circuit **153** operatively coupled to a communications antenna **151** and a locating antenna **159**. The processing circuit **153** may include a processor and a memory. The memory may contain one or more instructions that, when executed by the processor, cause the processing circuit **153** to perform one or more of the functions described herein. The processing circuit **153** may be further operatively coupled to one or more chassis sensors, such as chassis sensors **154** depicted in FIG. 1. The processing circuit **153** may be further coupled to one or more implement sensors, such as implement sensors **155** depicted in FIG. 5.

In some embodiments, the controller **150** is positioned onboard the service vehicle **100**. In some embodiments, the controller **150** is a cloud-based controller positioned remotely from the service vehicle **100**. In some embodiments, the controller **150** includes multiple controllers cooperating to provide the functionality of the controller **150** described herein. By way of example, the controller **150** may represent a first controller positioned onboard the service vehicle **100** and a second cloud-based controller in communication with one another.

In some embodiments, the communications antenna **151** may be configured to communicate with one or more other devices. The communications antenna **151** may be configured to receive location data from one or more external vehicles, such as one or more lift devices **700** depicted in FIG. 7. The communications antenna **151** may be configured to receive data regarding the battery life status (e.g., a charge

level) of the one or more lift devices **700**. The communications antenna **151** may be configured to receive location data from one or more charging systems for charging the service vehicle **100**. In some embodiments, the communications antenna **151** is configured to communicate using a wireless communication protocol, including but not limited to, Wi-Fi (e.g. 802.11x), Wi-Max, cellular (e.g. 3G, 4G, LTE, CDMA, etc.), LoRa, Zigbee, Zigbee Pro, Bluetooth, Bluetooth Low Energy (BLE), Near Field Communication (NFC), Z-Wave, 6LoWPAN, Thread, RFID, and other applicable wireless protocols. In some embodiments, the communications antenna **151** may communicate with a network (e.g. via Wi-Fi) of a job site, such as a job site **790** depicted in FIG. 7. In some embodiments, the communications antenna **151** may communicate with a network of the job site **790** to access a central server, such as a cloud based server, as will be described in more detail. In other embodiments, the communications antenna **151** may communicate with a local communications hub or bridge, such as a communications hub associated with one or more lift devices. In still other embodiments, the communications antenna **151** may be configured to allow for the service vehicle **100** to communicate directly with a central or cloud-based server (e.g. via a cellular connection). In some embodiments, the communications antenna **151** may be used to communicate with a user device capable of remotely controlling the service vehicle **100**. Example user devices capable of remotely controlling the service vehicle **100** may include dedicated remote controls, smart phones, tablet computers, laptop computers, or any other user device capable of interfacing with the controller **150**.

In some embodiments, the locating antenna **159** (e.g., a location sensor) may be configured to provide location data to the controller **150** of the service vehicle **100**. In some embodiments, the locating antenna **159** may be a Global Positioning Satellite (GPS) antenna for receiving locations data from a number of GPS satellites. In other embodiments, the locating antenna **159** may be configured to receive one or more differential GPS signals for determining a location of the service vehicle **100**. In still other embodiments, the locating antenna **159** may be configured to receive one or more inputs from a local positioning system. For example, the locating antenna **159** may be configured to receive data from an installed boundary system (such as a buried cable, or placed transmitters). Additional local positioning data may come from an aerial transmitter, such as on a rooftop or a drone. Further local positioning data may include optical boundary data, magnetic boundary data, etc. In still further embodiments, the locating antenna **159** may relay data to other service vehicles, or supervisory controllers to allow for a position of the service vehicle **100** to be monitored. However, in some embodiments, the location data of the service vehicle **100** may be communicated via the communications antenna **151**.

In some embodiments, the implement sensors **155** may be positioned around the surface of the charging implement **170** as depicted in FIG. 1, as well as in other locations as needed for a given configuration. The implement sensors **155** may provide implement position data indicating a position of the charging implement **170** relative to a portion of another vehicle or charging implement (e.g., the antenna coil **712** of the lift device **710**, an antenna coil that provides electrical energy to recharge the service vehicle **100**, etc.). The implement sensors **155** may be all of the same type, or may be a combination of different sensor types. The implement sensors **155** may include object detection sensors, such as infrared (IR), LIDAR, RADAR, Time-of-Flight (ToF),

CCD, CMOS, Ultrasonic, Sonar, or other sensors configured to detect objects. The implement sensors **155** may be used to detect a charging pad underneath a lift device. The implement sensors **155** may be used to detect a charging pad to charge the service vehicle. The implement sensors **155** may provide data to the controller **150**, which can process the data to perform functions as will be described in more detail below.

In some embodiments, the chassis sensors **154** on the service vehicle **100** may be positioned around the service vehicle **100** as depicted in FIG. 1, as well as in other locations as needed for a given configuration. The chassis sensors **154** may be all of the same type, or may be combination of different sensor types. The chassis sensors **154** may include object detection sensors, such as infrared (IR), LIDAR, RADAR, Time-of-Flight (ToF), CCD, CMOS, Ultrasonic, Sonar, or other sensors configured to detect objects. The chassis sensors **154** may be used to detect objects, to map the job site **790**, or to assist in guidance of the service vehicle **100**. Further sensors may include moisture sensors, rain sensors, air quality sensors, magnetic field sensors (e.g. compass), temperature sensors, digital imaging sensors, motion detection sensors, rotation sensors, gyroscopes, chemical detection sensors, and the like. In some embodiments, the chassis sensors **154** are coupled to the controller **150**, and are used to provide data to the controller **150**, which can process the data to perform functions as will be described in more detail below.

Referring now to FIG. 7, a process for wirelessly charging one or more lift devices **700** using a service vehicle, such as the service vehicle **100** depicted in FIG. 1, is shown according to one embodiment. The one or more lift devices **700** may include a lift device **710**, a second lift device **720**, and a third lift device **730**. Although shown as telescopic boom lift, the one or more lift devices **700** can each be a variety of different lift devices, including a scissor lift, telehandler, electric scissor lift, forklift, or other suitable devices that include one or more battery-operated or electrical components.

In some embodiments, a controller, such as the controller **150** depicted in FIG. 1 and described in further detail regarding FIG. 6, may be configured to use data received from the communications antenna **151** and/or the locating antenna **159**, to determine the location of each of the one or more lift devices **700**, the state of the battery life of each of the one or more lift devices **700**, and define an order of priority of charging stops via the processing circuit **153**. The processing circuit **153** may compile the data and provide the controller **150** with commands to communicate to the service vehicle **100** for autonomously driving into proximity with one of the one or more lift devices **700** to begin the process for wirelessly charging one of the one or more lift devices **700**. The controller **150** may receive data regarding the surrounding environment of the service vehicle **100** from the chassis sensors **154**. The processing circuit **153** may compile the data and provide additional commands to the service vehicle **100** to adjust course to one of the one or more lift devices **700** based on the data regarding the surrounding environment. For example, the chassis sensors **154** may communicate data regarding various obstructions on the job site **790** to avoid on route to one of the one or more lift devices **700**. The controller **150** may receive data regarding the surrounding environment of the charging implement **170** from the implement sensors **155**. The processing circuit **153** may compile that data and provide commands to the boom assembly **160** to articulate the position of the charging implement **170** to reach a charging area of one of the one or

more lift devices **700**, as described in further detail below. As shown, the service vehicle **100** may interact with a lift device **710** to charge a battery, such as a battery **714** depicted in FIG. **8**, of the lift device **710**. In some embodiments, the service vehicle may be commanded by the controller **150** to drive into proximity of two of the one or more lift devices **700** and charge both of the lift devices simultaneously with the boom assembly **160** and a secondary boom assembly, such as the secondary boom assembly **460** depicted in FIG. **4**.

Referring now to FIG. **8**, a process for wirelessly charging a lift device, such as the lift device **710** depicted in FIG. **7**, is shown according to one embodiment. The wireless charging process generally includes a service vehicle, such as the service vehicle **100** depicted in FIG. **1**, interacting with the lift device **710** as depicted in FIG. **7**, to discharge an electric charge through the charging pad **172** of the service vehicle **100**. To interact with the service vehicle **100**, the lift device **710** includes an antenna coil **712** that is electrically coupled with the battery **714** of the lift device **710**. The antenna coil **712**, like the induction coil **114** depicted in FIG. **5**, may be formed of copper wire that includes a series of turns. When the antenna coil **712** is positioned within a magnetic field, a current is generated within the antenna coil **712** that can then be provided to the battery **714** to help charge the battery **714**. Accordingly, to better position the antenna coil **712** within magnetic fields (e.g., such as the magnetic field created by the induction coil **114**), the antenna coil **712** is positioned at or near a base of a chassis **716** of the lift device **710**. Accordingly, the antenna coil **712** will be positioned at or near an absolute bottom of the chassis **716** of the lift device **710**.

In some embodiments, a service vehicle, such as the service vehicle **100** depicted in FIG. **1**, drives into proximity with the lift device **710**. As shown, the service vehicle **100** may then extend and/or pivot a boom assembly, such as the boom assembly **160** depicted in FIG. **1**, until the antenna coil **712** is positioned directly or approximately directly above the induction coil **114** of a charging pad, such as the charging pad **172** depicted in FIG. **1**, and the charging area **116** of the charging pad **172**, more generally. Although shown centered above the charging area **116** and the induction coil **114**, certain versions of the lift device **710** may have an antenna coil **712** offset to a different side of the lift device **710**. For example, in some embodiments, the antenna coil **712** is offset to one of the corners of the chassis **716**. Such a configuration may help facilitate the autonomous positioning of the service vehicle **100** and the autonomous positioning of the charging pad **172** when the job site **790** is crowded or where debris or other obstructions exist on the ground between the service vehicle **100** and the lift device **710**, given the limited reach of the boom assembly **160**. In other examples, the lift device **710** includes an indicator **715** (e.g., a light) on the chassis **716** or a lift device housing, such as a lift device housing **718** depicted in FIG. **7**, that illuminates when the antenna coil **712** is positioned within the charging area **116** and the battery **714** is receiving power (i.e., the battery is charging).

In some embodiments, with the lift device **710** positioned so that the antenna coil **712** is above the induction coil **114** and the charging area **116**, the antenna coil **712** is positioned within a magnetic field **800** created by the current passing through the induction coil **114**. The antenna coil **712** within the magnetic field **800** generates a current within the antenna coil **712**, which is then passed upwardly, to the battery **714** of the lift device **710** to charge the battery **714**. In some

examples, the indicator **715** can provide a visual indication that charging is complete, or that charging has reached a threshold level.

Although depicted as an induction coil **114**, various other types of wireless charging mechanisms can be used. For example, magnetic resonance charging, electric field coupling, or radio reception can be used in lieu of magnetic induction. While operationally different, the structure for each different type of wireless charging mechanism described above can be considered encompassed within the term “induction coil.”

Referring again to FIG. **7**, when a desired charge level has been reached, the controller of the service vehicle **100** can drive the service vehicle **100** and/or otherwise move the boom assembly **160** and charging pad **172** of the service vehicle **100**, away from the lift device **710**. The service vehicle **100** may then move to the next one of the one or more lift devices **700** on the order of priority, such as the lift device **720** or the lift device **730**, to complete a similar charging process.

In some embodiments, the service vehicle **100** may be further configured to autonomously recharge itself. The controller **150** may be configured to use data received from the electrical cabinet **110** of the service vehicle **100** to determine when the battery life (e.g., charge level, state of charge, etc.) of the battery assembly **112** of the service vehicle **100** is low (e.g., below a predetermined threshold state of charge). The controller **150** may use data received from the communications antenna **151** and/or the locating antenna **159** to determine where the service vehicle must go to engage a charging system and recharge the battery assembly **112**. In some embodiments, the charging pad **172** may be further configured to receive an electrical charge. The inverters received in the electrical cabinet **110** may transition alternating current electricity received by the charging pad **172** into direct current electricity for storage in one or more of the batteries in the battery assembly **112**. In some embodiments, one or more of the batteries in the battery assembly **112** may be exchanged with new batteries that may be delivered to the job site **790** from a remote charging station. In some embodiments, one or more of the batteries in the battery assembly **112** may be charged by receiving an electricity through a charging port **180** depicted in FIG. **1**, the charging port **180** coupled to the electrical cabinet **110**.

Using the above described service vehicles and methods, a jobsite can incorporate a wireless charging unit that can help to continuously charge lift devices and other equipment, according to one embodiment. The autonomous service vehicle can create a faster and more efficient way to charge devices remotely, which helps to ensure that devices at a jobsite are operable beyond the life of a single charge of a battery. While conventional equipment is typically only able to operate for as long as a single charge of a battery lasts, the service vehicle disclosed herein permits for extended use of equipment.

As utilized herein with respect to numerical ranges, the terms “approximately,” “about,” “substantially,” and similar terms generally mean $\pm 10\%$ of the disclosed values. When the terms “approximately,” “about,” “substantially,” and similar terms are applied to a structural feature (e.g., to describe its shape, size, orientation, direction, etc.), these terms are meant to cover minor variations in structure that may result from, for example, the manufacturing or assembly process and are intended to have a broad meaning in harmony with the common and accepted usage by those of ordinary skill in the art to which the subject matter of this disclosure pertains. Accordingly, these terms should be

interpreted as indicating that insubstantial or inconsequential modifications or alterations of the subject matter described and claimed are considered to be within the scope of the disclosure as recited in the appended claims.

It should be noted that the term “exemplary” and variations thereof, as used herein to describe various embodiments, are intended to indicate that such embodiments are possible examples, representations, or illustrations of possible embodiments (and such terms are not intended to connote that such embodiments are necessarily extraordinary or superlative examples).

The term “coupled” and variations thereof, as used herein, means the joining of two members directly or indirectly to one another. Such joining may be stationary (e.g., permanent or fixed) or moveable (e.g., removable or releasable). Such joining may be achieved with the two members coupled directly to each other, with the two members coupled to each other using a separate intervening member and any additional intermediate members coupled with one another, or with the two members coupled to each other using an intervening member that is integrally formed as a single unitary body with one of the two members. If “coupled” or variations thereof are modified by an additional term (e.g., directly coupled), the generic definition of “coupled” provided above is modified by the plain language meaning of the additional term (e.g., “directly coupled” means the joining of two members without any separate intervening member), resulting in a narrower definition than the generic definition of “coupled” provided above. Such coupling may be mechanical, electrical, or fluidic.

References herein to the positions of elements (e.g., “top,” “bottom,” “above,” “below”) are merely used to describe the orientation of various elements in the FIGURES. It should be noted that the orientation of various elements may differ according to other exemplary embodiments, and that such variations are intended to be encompassed by the present disclosure.

The hardware and data processing components used to implement the various processes, operations, illustrative logics, logical blocks, modules and circuits described in connection with the embodiments disclosed herein may be implemented or performed with a general purpose single- or multi-chip processor, a digital signal processor (DSP), an application specific integrated circuit (ASIC), a field programmable gate array (FPGA), or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A general purpose processor may be a microprocessor, or, any conventional processor, controller, microcontroller, or state machine. A processor also may be implemented as a combination of computing devices, such as a combination of a DSP and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration. In some embodiments, particular processes and methods may be performed by circuitry that is specific to a given function. The memory (e.g., memory, memory unit, storage device) may include one or more devices (e.g., RAM, ROM, Flash memory, hard disk storage) for storing data and/or computer code for completing or facilitating the various processes, layers and modules described in the present disclosure. The memory may be or include volatile memory or non-volatile memory, and may include database components, object code components, script components, or any other type of information structure for supporting the various activities and information structures described in the present disclosure. According to

an exemplary embodiment, the memory is communicably connected to the processor via a processing circuit and includes computer code for executing (e.g., by the processing circuit or the processor) the one or more processes described herein.

The present disclosure contemplates methods, systems and program products on any machine-readable media for accomplishing various operations. The embodiments of the present disclosure may be implemented using existing computer processors, or by a special purpose computer processor for an appropriate system, incorporated for this or another purpose, or by a hardwired system. Embodiments within the scope of the present disclosure include program products comprising machine-readable media for carrying or having machine-executable instructions or data structures stored thereon. Such machine-readable media can be any available media that can be accessed by a general purpose or special purpose computer or other machine with a processor. By way of example, such machine-readable media can comprise RAM, ROM, EPROM, EEPROM, or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other medium which can be used to carry or store desired program code in the form of machine-executable instructions or data structures and which can be accessed by a general purpose or special purpose computer or other machine with a processor. Combinations of the above are also included within the scope of machine-readable media. Machine-executable instructions include, for example, instructions and data which cause a general purpose computer, special purpose computer, or special purpose processing machines to perform a certain function or group of functions.

Although the figures and description may illustrate a specific order of method steps, the order of such steps may differ from what is depicted and described, unless specified differently above. Also, two or more steps may be performed concurrently or with partial concurrence, unless specified differently above. Such variation may depend, for example, on the software and hardware systems chosen and on designer choice. All such variations are within the scope of the disclosure. Likewise, software implementations of the described methods could be accomplished with standard programming techniques with rule-based logic and other logic to accomplish the various connection steps, processing steps, comparison steps, and decision steps.

It is important to note that the construction and arrangement of the service vehicle **100** as shown in the various exemplary embodiments is illustrative only. Additionally, any element disclosed in one embodiment may be incorporated or utilized with any other embodiment disclosed herein. For example, the boom assembly **460** of the exemplary embodiment shown in at least FIG. **4** may be incorporated in the service vehicle **100** of the exemplary embodiment shown in at least FIG. **1**. Although only one example of an element from one embodiment that can be incorporated or utilized in another embodiment has been described above, it should be appreciated that other elements of the various embodiments may be incorporated or utilized with any of the other embodiments disclosed herein.

What is claimed is:

1. A service vehicle, comprising:

a chassis;

a plurality of tractive elements coupled to the chassis;

a cabinet containing a battery pack and extending above the chassis;

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a wireless charging interface configured to wirelessly transfer energy from the battery pack to a recipient vehicle; and
 a boom assembly coupling the wireless charging interface to the chassis and configured to move the wireless charging interface relative to the chassis,
 wherein the wireless charging interface is repositionable between a storage position and an active position, wherein the wireless charging interface is located farther from the chassis in the active position than in the storage position, wherein the boom assembly extends away from the chassis in a first direction when the wireless charging interface is in the active position, wherein the boom assembly extends above the chassis and along a side of the cabinet when the wireless charging interface is in the storage position, wherein the wireless charging interface faces upward and away from the chassis when in the active position, and wherein the wireless charging interface faces in a second direction opposite the first direction when in the storage position.

2. The service vehicle of claim 1, wherein the boom assembly includes a base boom section coupled to the chassis and a fly boom section coupled to the wireless

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charging interface, and wherein the fly boom section is slidably coupled to the base boom section.

3. The service vehicle of claim 2, wherein the base boom section is configured to rotate relative to the chassis about a substantially vertical axis.

4. The service vehicle of claim 3, wherein the base boom section is configured to rotate relative to the chassis about a substantially horizontal axis.

5. The service vehicle of claim 2, wherein the fly boom section is translatable relative to the base boom section between an extended position and a retracted position, and wherein the fly boom section is pivotally coupled to the base boom section such that the fly boom section is rotatable relative to the base boom section about an axis of rotation at least when the fly boom section is in the extended position.

6. The service vehicle of claim 1, wherein the wireless charging interface extends above the cabinet when in the storage position.

7. The service vehicle of claim 1, wherein the tractive elements are configured to propel the service vehicle in a forward direction, and wherein the side of the cabinet is a front side of the cabinet.

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